

大学物理公式合集:

1. 毕-萨定律: $d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \vec{r}}{r^3}$

2. 磁偶极矩: $\vec{P}_m = I \cdot S \vec{n}$ (S 为 I 围住的区域)

3. 安培环路定理: $\oint \vec{B} d\vec{l} = \mu_0 \sum I_i$

4. 高斯定理: $\oint_S \vec{B} d\vec{S} = 0$

5. 两长直导线单位长度受力: $F = \frac{\mu_0 I_1 I_2}{2\pi a}$ (同向)

6. 载流线圈受力力矩 $\vec{M} = \vec{P}_m \times \vec{B} = I S \vec{n} \times \vec{B}$

7. 有介质时 磁化强度 $\vec{M}$, $\oint_L \vec{M} d\vec{l} = \sum I'$

安培: $\oint_L \vec{H} d\vec{l} = \sum I_i$

8. $\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M}$ $\vec{B} = \mu_0 \mu_r \vec{H} = \mu \vec{H}$

磁场强度 $\vec{H} = \frac{\vec{B}}{\mu}$

磁化强度 $\vec{M} = (\mu_r - 1) \vec{H}$, 磁化率 χ_m ($\mu_r = 1 + \chi_m$)

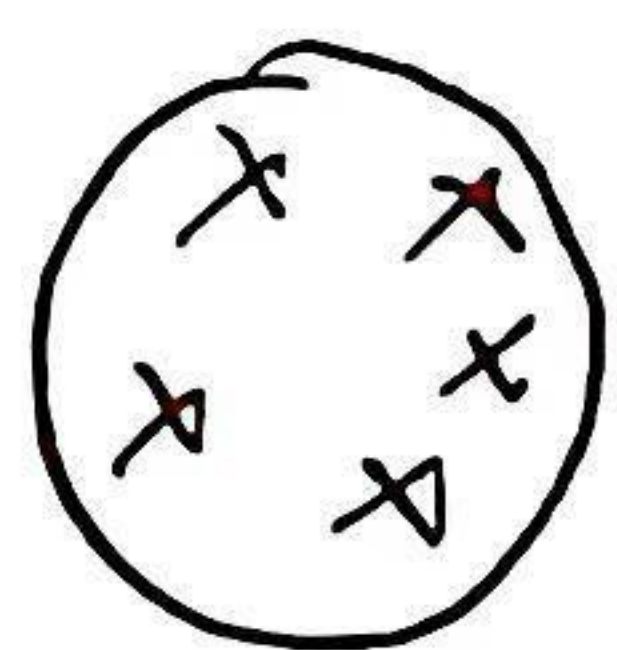
9. 感应电动势 $\varepsilon_i = -\frac{d\phi}{dt}$ ($\varepsilon_i = -N \frac{d\phi}{dt}$)

$I = -\frac{N}{R} \frac{d\phi}{dt}$ $Q = -\frac{N}{R} (\phi_1 - \phi_2)$

10. 动生电动势 $\varepsilon_t = \int_L (\vec{v} \times \vec{B}) d\vec{l}$

匀 ω 转动 $\varepsilon_i = -\frac{1}{2} B \omega l^2$. 圆盘: $\varepsilon_i = -\frac{1}{2} B \omega R^2$

11. 感生电场强度 $\oint_L \vec{E}_i d\vec{l} = -\int \frac{\partial \vec{B}}{\partial t} d\vec{S}$ ★★



$$\begin{cases} r < R \text{ 时 } \vec{E}_i = \frac{r}{2} \frac{\partial B}{\partial t} \text{ (逆正)} \\ r > R \text{ 时 } \vec{E}_i = \frac{R^2}{2r} \frac{\partial B}{\partial t} \end{cases}$$

12. 自感 $L = \frac{\psi}{I}$ 自感电动势 $\varepsilon_L = -\left(L \frac{dI}{dt} + I \frac{dL}{dt}\right)$

13. 互感 $M = \frac{\psi}{I}$ 互感电动势 $\varepsilon_m = -M \frac{dI}{dt}$ (M 为常数)

14. 振动 (谐振动) $\frac{d^2 x}{dt^2} + \omega^2 x = 0 \Rightarrow x = A \cos(\omega t + \varphi)$ $\omega = \sqrt{\frac{k}{m}}$

$v = \omega A \cos(\omega t + \varphi + \frac{\pi}{2})$

$a = -\omega^2 A \cos(\omega t + \varphi + \pi)$

$A = \sqrt{x_0^2 + \left(\frac{v_0}{\omega}\right)^2}$ $\tan \varphi = -\frac{v_0}{x_0 \omega}$

15. 谐振能量 $E = E_k + E_p = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \frac{1}{2}kA^2$ 此消彼涨, 相互转换

$$(E_k = E_p = \frac{1}{4}kA^2)$$

16. 振动合成 (同向, 同频) $A^2 = A_1^2 + A_2^2 + 2A_1A_2\cos\Delta\varphi$ $\tan\varphi = \frac{A_1\sin\varphi_1 + A_2\sin\varphi_2}{A_1\cos\varphi_1 + A_2\cos\varphi_2}$

(同向, 不同频) $\rightarrow$ 拍: $\nu = \nu_2 - \nu_1$

(垂直, 同频) $\Delta\varphi = \frac{\pi}{2}$ (右旋顺正) $\Delta\varphi = \frac{3\pi}{2}$ (左旋逆)

17. 共振 $\omega = \omega_0 = \sqrt{\omega_0^2 - 2\beta}$ (2β 为 $\frac{dx}{dt}$ 系数) ω_0^2 为 ω 系数

18. 波函数 $y = A \cos[\omega(t - \frac{x}{u}) + \varphi]$

19. 波传播中 $E_k = E_p = \frac{1}{2}E_{\text{总}}$. 数值相同. 大小相位相同 (机械能不守恒)

平衡位置 $E_{k\max}, E_{p\max}$

20. 平均能量密度 $\bar{w} = \frac{1}{2}\rho A^2\omega^2$. 能流 $P = wu\Delta S$. 能流密度 $\bar{S} = wu$

平均能流密度 (波强) $I = \bar{w}u = \frac{1}{2}\rho A^2\omega^2u \propto A^2$ ($I \propto A^2$)

21. 干涉条件: 同频, 方向相同, 相位差恒定

22. 干涉 $\begin{cases} A^2 = A_1^2 + A_2^2 + 2A_1A_2\cos\Delta\varphi \\ \Delta\varphi = (\varphi_2 - \varphi_1) - \frac{2\pi}{\lambda}(r_2 - r_1) \end{cases}$ $I = I_1 + I_2 + 2\sqrt{I_1I_2}\cos\Delta\varphi$

23. 驻波 $y = y_1 + y_2 = 2A \cos \frac{\omega}{u}x \cos \omega t$ $\begin{cases} \text{波节 } x = \pm(2k+1)\frac{\lambda}{4} \\ \text{波腹 } x = \pm k\frac{\lambda}{2} \end{cases} \begin{cases} \text{之间 } \Delta\varphi = 0 \\ \text{两侧 } \Delta\varphi = \pi \end{cases}$
相邻波节: $\frac{\lambda}{2}$ 相邻节腹: $\frac{\lambda}{4}$

位移 $\max \rightarrow$ 势能 $\max$; 平衡位置 $\rightarrow$ 动能 $\max$

24. 半波损失 (疏 $\rightarrow$ 密) $\Delta\varphi = \pi$. $\Delta x = \frac{\lambda}{2}$

25. 多普勒效应 $\nu_r = \frac{u - \nu_r}{u - \nu_s} \nu_s$ $\begin{cases} \text{靠近 } u \uparrow \\ \text{远离 } u \downarrow \end{cases}$

26. 电磁波条件: $\vec{E}, \vec{H}$ 同步变化, 同相位 $\nabla \cdot \vec{E} = \rho/\epsilon_0$ $\nabla \cdot \vec{H} = 0$
方向 $\vec{u} \parallel \vec{E} \times \vec{H}$ ($B = \sqrt{\epsilon_0\mu_0}E$)

27. 能流密度 $S = wu = EH$ 平均能流密度 $\bar{S} = \frac{1}{2}E_m H_m$

能量密度 $w = \frac{1}{2}EH$

光学

15. 28. $\delta = n_2 l_2 - n_1 l_1$

相位差 $\Delta\varphi = \frac{2\pi}{\lambda} \delta$ 相干叠加 $I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \Delta\varphi$

29. 双缝干涉 $d \sin \theta = \begin{cases} k\lambda & k=0,1,2\dots \text{明} \\ (2k+1)\frac{\lambda}{2} & \text{暗} \end{cases}$

间距 $\Delta x = \frac{D}{d} \lambda \rightarrow \lambda = \frac{\Delta x d}{D}$

(放入水中 $\Delta x' = \frac{D \lambda}{n d}$)

30. 光强 $I = 4I_0 \cos^2 \beta$. $\beta = \frac{\pi d \sin \theta}{\lambda}$

31. 洛埃镜 $\delta + \frac{\lambda}{2} = \begin{cases} k\lambda & \text{明} \\ (2k+1)\frac{\lambda}{2} & \text{暗} \end{cases}$

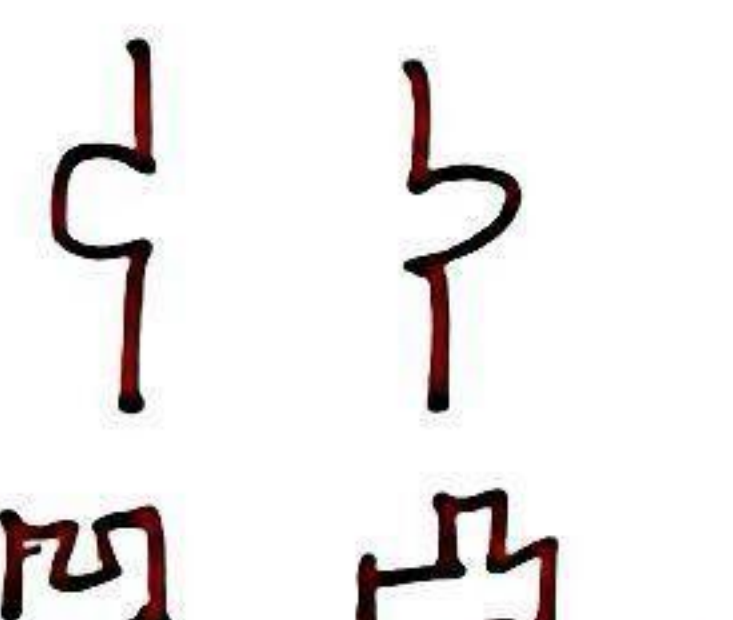
32. 等倾干涉 $\delta = 2d \sqrt{n^2 - n^2 \sin^2 i} = \begin{cases} k\lambda & \text{明 } k=1,2\dots \\ (2k+1)\frac{\lambda}{2} & \text{暗 } k=0,1,2\dots \end{cases}$ 中间级别最高 $\begin{cases} d \uparrow \text{外冒} \\ d \downarrow \text{右没} \end{cases}$

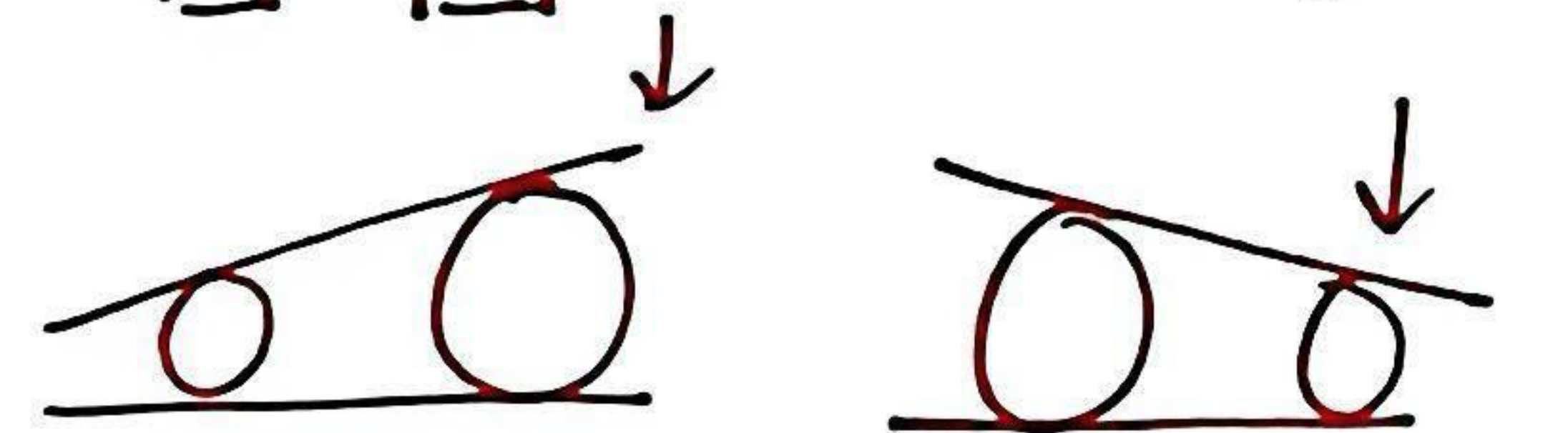
33. 附加 $\delta = \frac{bd}{2R}$ 极限宽度 $b_0 = \frac{R}{d} \lambda$

34. 劈尖干涉 (垂直入射) $2d + \frac{\lambda}{2} = \begin{cases} k\lambda & \text{明} \\ (2k+1)\frac{\lambda}{2} & \text{暗} \end{cases}$

$\Delta d = \frac{\lambda}{2} (\frac{\lambda}{2n})$

$L = \frac{\lambda}{2 \sin \theta}$

35.  $a \sin \theta = h$
 $\lambda \sin \theta = \frac{\lambda}{2} = \Delta d$

36.  $\begin{cases} \theta \downarrow \text{变密, 右小} \\ \theta \uparrow \text{变疏, 右大} \end{cases}$

37. 牛顿环. $2d + \frac{\lambda}{2} = \begin{cases} k\lambda \\ (2k+1)\frac{\lambda}{2} \end{cases}$ (暗)

中间一定暗. 暗纹半径 $r_k = \sqrt{k R \lambda}$ 上移 d_0 : $r = \sqrt{k(k\lambda - 2d_0)}$

38. 迈克尔逊: $2\Delta d = \delta = N\lambda$ (M_2 移动 $3\Delta d$)
 $\Delta d = N \frac{\lambda}{2}$

39. 衍射 $I_\theta = I_0 \left(\frac{\sin \alpha}{\alpha} \right)^2$ $\alpha = \frac{\pi a \sin \theta}{\lambda}$

40. 中央主极大 $\Delta x = 2f \frac{\lambda}{a}$ $\theta_1 = \frac{\lambda}{a}$ (装置放入水中. 收缩. 亮度变小)

41. 衍射极小: $a \sin \theta = k\lambda$ ($\sin \alpha = 0$) 衍射次极大 $a \sin \theta = \pm (2k+1) \frac{\lambda}{2}$

42. 半波带 $AC = a \sin \theta = \begin{cases} 2k \cdot \frac{\lambda}{2} & \text{暗} \\ (2k+1) \frac{\lambda}{2} & \text{明} \end{cases}$

43. 斜刀射 $d(\sin \alpha \mp \sin \theta) = \begin{cases} \pm k \lambda \\ \pm (2k+1) \frac{\lambda}{2} \end{cases}$
同 + 异 -

44. 双缝衍射 $I_\theta = I_0 \left(\frac{\sin \alpha}{\alpha} \right)^2 \cos^2 \beta$

极小: $d \sin \theta = (2k+1) \frac{\lambda}{2}$ ($a \sin \theta = k \lambda$)

极小间主极大: $d \sin \theta = k \lambda \longrightarrow \text{缺级 } k' = k \frac{d}{a}$

45. 光栅常数 $d = a + b$

$I_\theta = I_0 \left(\frac{\sin \alpha}{\alpha} \right) \left(\frac{\sin N\beta}{\sin \beta} \right)^2$

主极大 $d \sin \theta = k \lambda$. 相邻主极大之间 $(N-2)$ 个次极大

干涉极小 $\sin N\beta = 0$ $d \sin \theta = \pm \left(k + \frac{m}{N} \right) \lambda$

46. $\frac{d}{\lambda} = k_{\max}$

47. 圆孔衍射 $\theta_1 = 1.22 \frac{\lambda}{D}$ (最小分辨角)

"爱里斑" $R = 1.22 f \frac{\lambda}{D}$ 分辨率 $R = \frac{1}{\Delta \varphi} = \frac{D}{1.22 \lambda}$

48. 布拉格公式 $2d \sin \theta = k \lambda$

线偏 $\downarrow$

49. 马吕斯定律 $I = I_0 \cos^2 \alpha$ (检偏转一次, 两 $I_{\max}$, 两消光)

50. 布儒斯特角: $i = i_B$ $\tan i_B = \frac{n_2}{n_1}$

51. 半波片: 圆偏 $\rightarrow$ 线偏

线偏 $\rightarrow$ 椭圆偏

52. 偏振光公式 $\Delta \varphi_{\text{总}} = (n_o - n_e) d \frac{2\pi}{\lambda} + \pi = \begin{cases} 2k\pi & \text{明} \\ (2k+1)\pi & \text{暗} \end{cases}$